Rational equations

1 Use technology to solve the following equations. Round your answer to two decimal places where necessary.

a $\dfrac{1}{x}=5$

b $-\dfrac{1}{x}=-5x$

c $\dfrac{1}{x-8}=-\dfrac{1}{x+1}$

d $\dfrac{1}{2x^2}=5$

e $\dfrac{8x-22}{x-3}=\dfrac{3x-25}{x-9}$

f $\dfrac{-5x^2-40x-81}{x^2+8x+16}=\dfrac{-5x-19}{x+4}$

g $\dfrac{10}{x^2+2x-15}=\dfrac{x^2+10}{5}$

h $\dfrac{x^2+x-12}{x-5}=\dfrac{2x+3}{x+4}$

2 Consider the equation $\dfrac{5}{x+3}+\dfrac{4}{x+4}=\dfrac{5}{x^2+7x+12}$.

a Solve for x.

b Explain whether your answer from part (a) is a valid solution.

3 For each of the following equations, determine whether $x=4$ is a valid solution, an extraneous solution, or not a solution at all:

a $\dfrac{x+3}{x-4}=0$

b $\dfrac{x-4}{x+3}=0$

c $\dfrac{(x-4)(x+3)}{x-4}=0$

d $\dfrac{x-4}{x-4}=0$

e $\dfrac{(x-4)^2}{x-4}=0$

f $\dfrac{x}{4}-\dfrac{4}{x}=0$

4 Find the valid solutions to the following equations:

a $\dfrac{1}{x-4}+\dfrac{1}{x+9}=0$

b $\dfrac{x}{x-2}+\dfrac{1}{x+2}=\dfrac{8}{x^2-4}$

c $\dfrac{9}{x+4}-\dfrac{5}{x-4}=\dfrac{2x}{x^2-16}$

d $\dfrac{2}{3x+5}+\dfrac{2}{3x-5}=\dfrac{5x}{9x^2-25}$

e $\dfrac{3}{x^2-2x}-\dfrac{5}{x^2-4}=0$

f $\dfrac{5}{x^2-2x}-\dfrac{3}{x^2-4x}=0$

g $1-\dfrac{7}{x}+\dfrac{10}{x^2}=0$

h $1+\dfrac{9}{x}=-\dfrac{20}{x^2}$

i $\dfrac{3x}{3-x}+\dfrac{3}{x}-5=0$

j $\dfrac{2x}{x-8}+\dfrac{6}{x+8}=\dfrac{4}{64-x^2}$

5 Find the valid solutions to the following equ[redacted]s:

a $2x^{-1}+5\left(2x-5\right)^{-1}=4\left(2x-5\right)^{-1}$

b $3x^{-1}+2\left(5x-4\right)^{-1}=5\left(5x-4\right)^{-1}$

c $x\left(x-9\right)^{-1}+x\left(x+9\right)^{-1}=32\left(x^2-81\right)^{-1}$

d $x\left(x-7\right)^{-1}+x\left(x+7\right)^{-1}=18\left(x^2-49\right)^{-1}$

6 Find all the complex solutions to $x^{-4}+8x^{-2}-9=0$.

Graphical solutions to rational equations

7 For each of the following functions, use technology to find:

i The number of x-intercepts.

ii The number of y-intercepts.

iii The number of horizontal asymptotes.

iv The number of vertical asymptotes.

a $f\left(x\right)=\dfrac{x-5}{x-3}$

b $f\left(x\right)=\dfrac{x^2+3x-10}{x+7}$

c $f\left(x\right)=\dfrac{x+2}{x^2+x-6}$

d $f\left(x\right)=\dfrac{x^2+5x-14}{x^2-3x-18}$

8 Sketch the graph of the following functions:

a $y=\dfrac{4\left(5-x\right)}{x-4}$

b $f\left(x\right)=\dfrac{x+5}{x-9}$

c $y=\dfrac{4}{\left(x-3\right)^2}$

d $y=\dfrac{x^2}{\left(x-6\right)\left(x+5\right)}$

e $y=\dfrac{-10\left(x+2\right)}{\left(x+5\right)\left(x-8\right)}$

f $f\left(x\right)=\dfrac{x-2}{x^2-14x+48}$

g $f\left(x\right)=\dfrac{x^2-4x-45}{x^2-6x-16}$

h $f\left(x\right)=\dfrac{x^2-x-6}{x+6}$

9 Consider the rational equation $\dfrac{3}{x+4}=-\dfrac{1}{x-3}$.

a Sketch the graph of $y=\dfrac{3}{x+4}$.

b Sketch the graph of $y=-\dfrac{1}{x-3}$ on the same coordinate plane.

c State the solution to the rational equation.

10 Consider the rational equation $\frac{4}{x-3} - \frac{6}{1-x} = 0$.

a Sketch the graph of $y = \frac{4}{x-3}$.

b Sketch the graph of $y = \frac{6}{1-x}$ on the same coordinate plane.

c Find the solution to the rational equation.